

## Lecture 5

# Managing files and folder permission

*CoSc2043 – Network and System Administration*

# Lecture Overview

- ◆ Types of Files (Regular files, directories)
- ◆ Character and block files
- ◆ File attributes and ownership
- ◆ Link Files (Hard links, symbolic links)
- ◆ Changing access permissions
- ◆ Special attributes
- ◆ Searching the filesystem

# Types of Files

Linux defines seven types of files.

| No | File type             | Symbol | Created by        | Removed by          |
|----|-----------------------|--------|-------------------|---------------------|
| 1  | Regular file          | -      | editors, cp, etc. | <b>rm</b>           |
| 2  | Directory             | d      | <b>mkdir</b>      | <b>rmdir, rm -r</b> |
| 3  | Character device file | c      | <b>mknod</b>      | <b>rm</b>           |
| 4  | Block device file     | b      | <b>mknod</b>      | <b>rm</b>           |
| 5  | Local domain socket   | s      | <b>socket(2)</b>  | <b>rm</b>           |
| 6  | Named pipe            | p      | <b>mknod</b>      | <b>rm</b>           |
| 7  | Symbolic link         | l      | <b>In -s</b>      | <b>rm</b>           |

# Types of Files (Cont'd)

- ◆ Regular Files :- Most files used directly by a human user are regular files. For example, executable files, text files, and image files are regular files.
- ◆ Directories:- organize folders and files into a hierarchical structure.
- ◆ Devices
  - Character Devices
  - Block Devices
- ◆ Sockets and Named Pipes
- ◆ Symbolic Links : is a pointer or an alias to another file.

# Files and Directories

◆ Everything is referenced via a file

◆ Directories

- List of files & inodes
- "." – Reference to the current directory
- ".." – Reference to parent directory
- Root (/) – "." and ".." are the same

◆ Inodes (index node) :- contains details about each file, such as the node, owner, file, location of file

# Standard directories and their contents

| Pathname       | Contents                                                     |
|----------------|--------------------------------------------------------------|
| /bin           | Commands needed for minimal system operability               |
| /boot          | Kernel and files needed to load the kernel                   |
| /dev           | Device entries for disks, printers, pseudo terminals, etc.   |
| /etc           | Critical startup and configuration files                     |
| /home          | Home directories for users                                   |
| /lib           | Libraries and parts of the C compiler                        |
| /media         | Mount points for filesystems on removable media              |
| /opt           | Optional application software packages (not yet widely used) |
| /proc          | Information about all running processes                      |
| /root          | Home directory of the superuser (often just /)               |
| /sbin          | Commands for booting, repairing, and recovering the system   |
| /tmp           | Temporary files that may disappear between reboots           |
| /usr           | Hierarchy of secondary files and commands                    |
| /usr/bin       | Most commands and executable files                           |
| /usr/include   | Header files for compiling C programs                        |
| /usr/lib       | Libraries; also, support files for standard programs         |
| /usr/local     | Local software (software you write or install)               |
| /usr/local/bin | Local executables                                            |
| /usr/local/etc | Local system configuration files and commands                |

# Standard directories and their contents (Cont'd)

| Pathname               | Contents                                                        |
|------------------------|-----------------------------------------------------------------|
| <b>/usr/local/bb</b>   | Local support files                                             |
| <b>/usr/local/sbin</b> | Statically linked local system maintenance commands             |
| <b>/usr/local/src</b>  | Source code for /usr/local/*                                    |
| <b>/usr/man</b>        | On-line manual pages                                            |
| <b>/usr/sbin</b>       | Less essential commands for system administration and repair    |
| <b>/usr/share</b>      | Items that might be common to multiple systems (read-only)      |
| <b>/usr/share/man</b>  | On-line manual pages                                            |
| <b>/usr/src</b>        | Source code for nonlocal software packages (not widely used)    |
| <b>/var</b>            | System-specific data and configuration files                    |
| <b>/var/adm</b>        | Varies: logs, system setup records, strange administrative bits |
| <b>/var/log</b>        | Various system log files                                        |
| <b>/var/spool</b>      | Spooling directories for printers, mail, etc.                   |
| <b>/var/tmp</b>        | More temporary space (preserved between reboots)                |

# Device Files

## ◆ Character Devices

- Transfer unit: byte
- Example: `/dev/console`

## ◆ Block Devices

- Transfer unit: Group of bytes (block)
- Examples: `/dev/hda`

## ◆ Device Numbers

- Major – Type of device
- Minor – Device number



# Device Files (Cont'd)

| Character Driver                                                                       | Block Driver                                                                                           |
|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Character devices are accessed as a stream of sequential data, one byte after another. | Block Device are hardware devices which randomly access fixed-sized chunks of data.                    |
| No buffering is required.                                                              | Accessed through a cache so buffering is required.                                                     |
| Seeking is not allowed.                                                                | Seeking is possible.                                                                                   |
| Managing char driver within a kernel require less care, preparation and work done.     | Managing block driver within a kernel requires more care, preparation and work done.                   |
| Character driver has only one position current one. It can't move back and forth.      | Block driver navigates back and forth between any location on media.                                   |
| The read() and write() calls do not return until the operation is complete.            | Reads and write done by buffer cache mechanism by bread(), bwrite().These request may be asynchronous. |
| The kernel doesn't have to provide an entire subsystem to the character device.        | The kernel does have to provide an entire subsystem to block device.                                   |
| sources for character devices are kept in .../kernel/chr_drv/                          | Sources for block devices are kept in .../kernel/blk_drv/                                              |
| Less complex.                                                                          | More complex.                                                                                          |
| Eg: Keyboard device, serial port.                                                      | Eg: Hard disk, Pen drive, Blue-rays.                                                                   |

# Sockets & Named Pipes

- ◆ Enables communication between processes

- ◆ Socket

- Processed must have a connection first
- Example: X Windows

- ◆ Named Pipe

- Communication between unrelated processes
- FIFO
- Not used very often

# Link Files

◆ Multiple names for same file

◆ Hard Link

- Pointer to Inode
- Can't cross partitions
- Equivalent to the original file reference
- File removed when all links deleted

◆ Symbolic (Soft) Links

- Pointer to file path name
- Dangling symlink – *Real* file which no longer exists

◆ `ln [-s] <real_file> <link_file>`

# Magic Numbers

- ◆ Byte pattern at beginning of file
- ◆ Patterns listed in file called **magic**  
RedHat: `/usr/share/magic`
- ◆ **file** – Tests a file to determine type
  - Filesystem Test
  - Magic Number Test
  - Language Test

# MS File/Directory Attributes



# UNIX/Linux File Attributes

| Inode                                                                                                        |
|--------------------------------------------------------------------------------------------------------------|
| Permissions                                                                                                  |
| Ownership                                                                                                    |
| Time Stamps <ul style="list-style-type: none"><li>♦ Change</li><li>♦ Modification</li><li>♦ Access</li></ul> |
| File Size                                                                                                    |
| Link Count                                                                                                   |
| Pointers to data                                                                                             |

# Viewing File Attributes

**ls -l**: Long listing (includes attributes)

- ◆ File Type
- ◆ File Size/Device #
- ◆ Permissions
- ◆ Modification Date
- ◆ Link Count
- ◆ File Name
- ◆ Ownership

**stat**: Lists all attributes

# File Type Attribute

```
# ls -ld /home /etc/passwd /dev/console
```

```
crw--w--w- 1 root    root      5,  1 Sep 27 11:27 /dev/console
-rw-r--r-- 1 root    root      559 Sep 22 13:14 /etc/passwd
drwxr-xr-x 3 root    root      0 Sep 26 10:42 /home
```

| File Type | Meaning          |
|-----------|------------------|
| -         | Regular File     |
| d         | Directory        |
| l         | Symbolic Link    |
| b         | Block Device     |
| c         | Character Device |
| p         | Named Pipe       |
| s         | Domain Socket    |



# Ownership

```
# ls -ld /home /etc/passwd /dev/console
crw--w--w-  1 root    root      5,   1 Sep 27 11:27 /dev/console
-rw-r--r--  1 root    root      559 Sep 22 13:14 /etc/passwd
drwxr-xr-x  3 root    root      0 Sep 26 10:42 /home
```

## ◆ User

- Owner of file
- User names/UIDs defined in `/etc/passwd`

## ◆ Group

- Organization of users accessing the file
- Group names/GIDs defined in `/etc/group`

# Permissions

```
# ls -ld /home /etc/passwd /dev/console
crw--w--w-  1 root    root      5,   1 Sep 27 11:27 /dev/console
-rw-r--r--  1 root    root      559 Sep 22 13:14 /etc/passwd
drwxr-xr-x  3 root    root       0 Sep 26 10:42 /home
```

| Operation | File               | Directory          |
|-----------|--------------------|--------------------|
| Read      | Read file          | List files         |
| Write     | Delete/Modify file | Create/Delete file |
| Execute   | Run program        | Access file        |

◆ 3 levels of access – Owner, Group, Other

# Changing Access

## ◆ Users & Groups

- `chown [-R] user file...`
- `chgrp [-R] group file`

## ◆ Permissions

- `chmod [-R] <op> file...`
- Numeric: `<op> = [#]###`
- Symbolic: `<op> = <who op perm>`
  - ◆ Who: (u)ser (g)roup (o)ther (a)ll
  - ◆ op: (+)add (-)remove (=)set
  - ◆ Perm: (r)ead (w)rite e(x)ecute

# Numeric/Symbolic Permissions

| Octal | Binary | Symbolic |
|-------|--------|----------|
| 0     | 000    | ---      |
| 1     | 001    | --x      |
| 2     | 010    | -w-      |
| 3     | 011    | -wx      |
| 4     | 100    | r--      |
| 5     | 101    | r-x      |
| 6     | 110    | rw-      |
| 7     | 111    | rwx      |

# Default Permissions

- ◆ **umask** Shell Environment Variable
- ◆ Defines permissions to remove

| Numeric | Binary | Effective Perms |
|---------|--------|-----------------|
| 0       | 000    | <b>rwX</b>      |
| 1       | 001    | <b>rw-</b>      |
| 2       | 010    | <b>r-w</b>      |
| 3       | 011    | <b>r--</b>      |
| 4       | 100    | <b>-wX</b>      |
| 5       | 101    | <b>-w-</b>      |
| 6       | 110    | <b>--X</b>      |
| 7       | 111    | <b>---</b>      |

# Special Attributes

## ◆ Setuid (SUID) Bit

- Run program with access of owner
- Symbolic: **s**
- Numeric: 4000

## ◆ Setgid (SGID) Bit

- Run program with access of owner group
- Symbolic: **s**
- Numeric: 2000

# Sticky Bit

## ◆ Purpose

- File: Force program to stay in RAM (obsolete)
- Directory: Cannot remove file unless you own the file or directory

◆ Symbolic: `t`

◆ Numeric: 1000

◆ Example: `/tmp`

# Microsoft Permissions

- ◆ Read
- ◆ Create
- ◆ Write
- ◆ Append
- ◆ Delete
- ◆ Execute
- ◆ Search
- ◆ Ownership
- ◆ Access Control

**GUI and Command Line (`cacls`) tools to manage**



# Searching the Filesystem

**find:** Command line search tool

- ◆ Searches through directory hierarchy
- ◆ Search by any combination of file names and attributes
- ◆ Display files or perform operations on them
- ◆ Examples:
  - `find /var -mtime -1`
  - `find / -name core -exec rm -f {} \;`